

Daniel David

Software Engineer | Specialized in Secure ML Pipelines & AI Engineering | Columbia University
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WORK EXPERIENCE

Course Assistant (COMS4995 ML Security) | Columbia University, NY | [Link](#) Jan. '26 – May '26

- Launched a new graduate-level course with Dr. Suman Jana focused on the security of ML and LLM frameworks.
- Mentoring students in the development of novel research projects while managing logistics and technical evaluations.
- Selected based on unique early industry expertise to bridge the gap between academic and practical research applications.

Software Engineer | Rhino Federated Computing | Remote | [Link](#) Dec. '24 – Jan. '26

- **Product Prototyping:** Engineered production-grade ML pipelines to be used on the company's GUI. Customized them rapidly based on the different clients' needs, particularly for the clients under the Eli Lilly TuneLab project. [Link](#).
- **System Reliability & Monitoring:** Set up real-time system monitoring using Grafana to track network health and manage latency issues during live distributed workgroups.
- **Encryption & Data Integrity Mechanisms:** Designed a secure way to transmit ML models using encryption and AWS Secrets Manager to release keys during runtime, adhering to strict client security requirements. [Link](#).
- **Documentation & Onboarding:** Authored the flagship technical documentation regarding GUI and SDK usage, serving as the primary onboarding document and additional ongoing technical guidance and support. [Link](#), [Link](#).
- **Customer Support:** Provided direct hands-on support for new enterprise clients and research collaborators with the platform.

Security Officer - Ministry of Foreign Affairs of Israel (Israeli Embassy in Washington D.C.) May '21 – Aug. '22

- Acted as diplomatic security officer and bodyguard for the Israeli ambassador and military attaché with strict confidentiality.

Combat Soldier - IDF, "Maglan" Unit Nov. '17 – Nov. '20

- Full combat service in a commando unit included 15 months of intense training, operational, and commanding experience.

EDUCATION

Columbia University - Bachelor of Arts - Computer Science. Sep. '22 – May '26

ML/AI track, Recipient of the "Renaissance" Honors Scholarship, Dean's List, PTK Honor Society.

TECHNICAL SKILLS

- **AI Agentic & RAG Engineering:** Designing multi-agent workflows and autonomous loops using OpenAI Agents SDK, CrewAI, and LangChain. Focused on integrating LLMs with structured databases to build reliable, tool-using agents. [Link](#)
- **Machine Learning:** Core expertise in building and training models using PyTorch and scikit-learn. Experienced in developing solutions for complex tasks. [Link](#)
- **Federated Learning & Security:** Professional expertise in decentralized learning using NVIDIA's NVFlare. Authored technical guides and developed AES+RSA encryption workflows with AWS Secrets Manager for key handling. [Link](#)
- **Full-Stack & Rapid Prototyping:** Building and deploying end-to-end AI applications using React, Node.js, and JavaScript. Experienced in rapid prototyping with Streamlit and Gradio and deploying scalable web services via Vercel and Supabase.
- **Databases & Data Engineering:** Proficient in SQL and PostgreSQL for production pipelines, with experience in Supabase for backend-as-a-service. Academic and practical exposure to NoSQL systems, including MongoDB and Neo4j.
- **Cloud & Infrastructure:** Extensive professional experience with containerization (Docker) while deploying federated workloads across AWS, GCP, and Azure. Utilized Grafana for real-time system monitoring.
- **Programming:** Strong professional experience with Python. Additional proficiency in C, C++, Java, and Assembly.

SELECTED PROJECTS

Clawdstrike (Backbay Industries) [Link](#)

Collaborated on an open-source project with over 270 stars. A novel security gateway for AI agents to solve runtime security and compliance bottlenecks for enterprise deployments. Integrating Rust-native boundaries with cryptographic attestation.

Rhino Tutorial #1 & #2– Basic Usage of Rhino FCP and Rhino's SDK for Federated Learning workflows [Link](#), [Link](#)

Created the flagship tutorial for the Rhino FCP to guide users through the fundamental steps of the platform's GUI.

Authored and maintained an advanced technical tutorial for the Rhino FCP's Python SDK.

Rhino's Model & Code Encryption for Secure Federated Computing Workflows [Link](#)

Developed a workflow to encrypt ML models and training logic in Rhino's Federated Computing Platform, with secure key handling via AWS Secrets Manager or JSON keyfiles, along with an article that now serves as an advanced reference guide.

Professional AI Representative [Link](#)

Developed an agent representing my professional persona 24/7, utilizing RAG with personal data and diverse range of tools.

Personal Interests: Innovation, Architecture, Long-Running and Endurance Sports, Skiing, Basketball.